CARDIA: Cardiovascular Risk Visualization & Multi-Vessel Stenosis AI Workstation

Tabular Multi-Target Prediction Integrated with 3D Anatomical Coronary Reference Visualization

CHALLENGE TRACK & SCOPE	DATASET PROVENANCE & IDENTIFIER	SUBMISSION DATE & STATUS
Track A: Cardiovascular Risk Visualization & Prediction	**UCI Extension of Z-Alizadeh Sani (#411)**	**October 2026 • Verified Build**

1. PROJECT OVERVIEW

1.1 Clinical Problem Statement: Coronary artery disease (CAD) represents the primary etiology of global cardiovascular morbidity and mortality. Definitive anatomical diagnosis requires invasive catheter coronary angiography (ICA) or coronary computed tomography angiography (CCTA). While ICA remains the gold standard for quantifying luminal narrowing, it carries procedural risks, institutional expense, and catheterization laboratory resource bottlenecks. Conversely, routine outpatient clinical workups capture detailed non-invasive data—patient demographics, resting 12-lead electrocardiography (ECG), blood biochemistry, and transthoracic echocardiography (TTE). However, conventional clinical scoring frameworks (such as Framingham or ASCVD calculators) aggregate these modalities into monolithic, non-anatomical risk indices that fail to communicate vessel-specific disease localization.

1.2 System Purpose & Key Contribution: CARDIA is an open-source clinical decision-support and visual exploration prototype engineered to bridge routine tabular clinical evaluation with multi-vessel anatomical risk communication. Rather than predicting only a monolithic binary CAD label, CARDIA evaluates four distinct targets confirmed via catheterization: overall CAD (`Cath`) alongside vessel-specific stenosis (≥50% luminal narrowing) across the three principal coronary arteries: the Left Anterior Descending (`LAD`), Left Circumflex (`LCX`), and Right Coronary Artery (`RCA`). These statistical predictions are mapped in real time to an interactive 3D WebGL anatomical coronary vascular tree, accompanied by patient-level feature attributions and transparent cross-validation performance metrics.

> **Anatomical Reference Standard & Scientific Scope**
> To maintain scientific transparency, the 3D heart and coronary meshes in CARDIA are standard reference geometries derived from the BodyParts3D digital atlas (DBCLS, Japan). They do **not** represent patient-specific CCTA reconstructions, fluoroscopy, or intravascular ultrasound. Vessel color tiers depict model-predicted tabular statistical probabilities (≥50% stenosis likelihood), never physical plaque geometry, calcification, or localized lesions. CARDIA is an educational research prototype and cannot replace formal angiography or clinical judgment.

2. DATASET PROVENANCE AND PREPROCESSING

2.1 Cohort Provenance: Models were trained and evaluated on the *Extension of Z-Alizadeh Sani Dataset*, archived in the UC Irvine Machine Learning Repository (UCI ID 411; DOI: **10.24432/C5461K**). Collected at Tehran Heart Center, the cohort comprises **303 consecutive patients** referred for diagnostic catheter angiography (303 rows × 59 raw columns).

2.2 Feature Audit & Clinical Domains: The dataset yields **54 clean predictor features** across six clinical domains:

Demographics (4): Age (observed 30–86 yrs, mean 58.9), Sex, Weight (kg), Length (cm).
Risk Factors & History (8): BMI, DM, HTN, Current Smoker, Ex-Smoker, FH, Obesity, DLP.
Comorbidities (5): CRF, CVA, Airway Disease, Thyroid Disease, CHF.
Bedside & Symptoms (9): BP, PR, Edema, Weak Pulse, Rales, Murmurs, Chest Pain, Dyspnea, NYHA Class.
Resting 12-Lead ECG (8): Q Wave, ST Elevation, ST Depression, T Inversion, LVH, Poor R Progression, BBB.
Biochemistry & Echo (17): FBS, CR, TG, LDL, HDL, BUN, ESR, HB, K, Na, WBC, Diff, PLT, EF, RWMA, VHD.

2.3 Preprocessing & Anti-Leakage Quarantine: Preprocessing pipeline steps applied: (1) Typographical correction: standardized `Sex` ('Fmale' corrected to 'Female'); (2) Constant feature drop: `Exertional CP` exhibited zero variance (100% 'N') and was dropped; (3) Data completeness: zero missing values across all 303 rows; (4) **Target Quarantine:** Invasive diagnostic catheterization outcomes `Cath`, `LAD`, `LCX`, and `RCA` were strictly segregated from the predictor matrix and barred from serving as cross-predictors. Inbound payloads containing target fields are rejected by the API with HTTP 400.

TARGET VARIABLE	CLINICAL DIAGNOSTIC SIGNIFICANCE	POSITIVE (STENOTIC/CAD)	NEGATIVE (NORMAL)	POSITIVE PREVALENCE
Cath (CAD)	Significant CAD in ≥1 major coronary vessel	216	87	71.29%
LAD	Left Anterior Descending luminal stenosis ≥50%	177	126	58.42%
LCX	Left Circumflex luminal stenosis ≥50%	119	184	39.27%
RCA	Right Coronary Artery luminal stenosis ≥50%	114	189	37.62%

2.4 Evidentiary Caveats: (1) *Row 94 Discrepancy:* Patient row 94 records `Cath`=Normal while `LAD`=Stenotic. Under formal clinical criteria, isolated ≥50% LAD stenosis constitutes CAD. This discrepancy was preserved intact to maintain forensic fidelity to the clinical registry. (2) *Referral Bias:* The high baseline disease prevalence (71.3% CAD) reflects tertiary hospital catheterization referrals and does not mirror asymptomatic general population screening.

3. METHODOLOGY & MACHINE LEARNING PIPELINE

3.1 Multi-Target Problem Formulation: CARDIA formulates risk estimation as four distinct supervised binary classification tasks over the clinical feature space $X \subset \mathbb{R}^{54}$:

$$f_{\text{Cath}}(x),\ f_{\text{LAD}}(x),\ f_{\text{LCX}}(x),\ f_{\text{RCA}}(x) \in [0, 1]$$

Independent estimators are utilized for each target. This avoids error propagation across vessels and allows algorithm selection tailored to the physiological profile of each vascular territory.

3.2 Candidate Model Families: Three distinct machine learning paradigms were trained and cross-validated for each target:

- **Regularized Logistic Regression (`LogisticRegression`):** L2-penalized linear estimator (C = 1.0, L-BFGS solver, max iterations 1000). Serves as an interpretable linear baseline.
- **Random Forest Classifier (`RandomForestClassifier`):** Non-parametric bagged ensemble of 200 decision trees (`n_estimators=200`, `max_depth=6`, `min_samples_split=4`, random seed 42) capable of capturing feature interactions without strong distributional assumptions.
- **Gradient Boosting Classifier (`GradientBoostingClassifier`):** Sequentially stage-wise additive boosted tree ensemble (`n_estimators=100`, `learning_rate=0.05`, `max_depth=3`, `subsample=0.8`, random seed 42) optimizing deviance loss.

3.3 Leakage-Proof Cross-Validation Architecture: To eliminate data leakage and optimistic bias:

- **Stratified 5-Fold Partitioning:** The 303 samples were split into 5 stratified folds for each target, preserving empirical positive/negative prevalence across all training and test splits.
- **Inside-Fold Pipeline Encapsulation:** Preprocessing was strictly bound inside scikit-learn `Pipeline` objects using `ColumnTransformer`. Numerical features (38 columns) were standardized via `StandardScaler`; categorical features (16 columns) were one-hot encoded via `OneHotEncoder(handle_unknown='ignore', sparse_output=False)`. Scaling parameters (mean, standard deviation) and categorical vocabularies were fitted solely on the 4 training folds and applied out-of-fold to the evaluation fold. No global transformation preceded splitting.

3.4 Model Selection Criterion: In clinical screening, false negatives (failing to identify an ischemic coronary stenosis) carry substantial risk of unheralded infarction, whereas false positives lead to non-invasive functional testing. Relying solely on classification accuracy distorts model ranking under class imbalance. Candidate models were selected by maximizing a composite clinical utility score:

$$\text{Score}_{\text{composite}} = 0.40 \times F_1 + 0.35 \times \text{ROC-AUC} + 0.25 \times \text{Recall}$$

This criterion enforces a balanced tradeoff between discriminative rank ordering (ROC-AUC), precision-recall balance (F_1), and high sensitivity (Recall) for ischemic lesion detection.

3.5 Feature Attribution & Explainability Methodology: To explain model behavior transparently without confounding statistical correlation with clinical causality:

- **Global Permutation Importance:** Evaluated across out-of-fold cross-validation sets using 10 random permutations. Features causing the largest reduction in scoring metrics when permuted were ranked highest.
- **Patient-Level Local Attributions:** For real-time inference on a specific patient payload, CARDIA computes local marginal contributions against the empirical background distribution. The top positive risk drivers (elevating probability) and top negative risk mitigators (lowering probability) are returned with baseline references.
- **Associational vs. Causal Boundary:** Documentation and UI components explicitly declare that feature attributions represent statistical model sensitivities within this retrospective cohort, not causal therapeutic targets. For example, while elevated ESR increases model-predicted risk, treating ESR directly without addressing underlying inflammatory or coronary disease does not imply clinical risk reduction.

3.6 Anatomical Vessel Colormap Projection: Predicted probabilities for the coronary arteries are mapped to 3D vascular branches using three intuitive visual triage tiers:

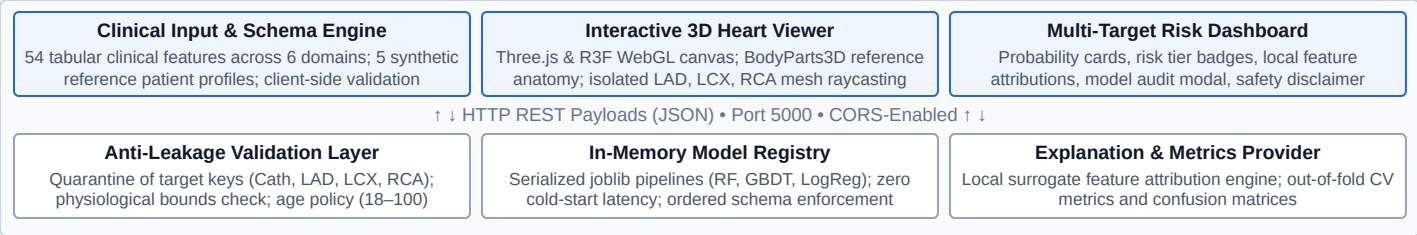
Coronary Vessel Probability Thresholds:
- LOW RISK (< 30%) : Emerald Green (`#10b981`) — Low statistical stenosis likelihood.
- MODERATE RISK (30%–60%) : Amber (`#f59e0b`) — Intermediate risk warranting surveillance.
- HIGH RISK (> 60%) : Crimson Red (`#ef4444`) — High likelihood of ≥50% stenosis.

Vessel-Specific Anatomical Scope:
- **LAD:** Anterior interventricular wall, septum, apex.
- **LCX:** Lateral left ventricle, posterior left ventricular wall.
- **RCA:** Inferior wall, right ventricle, sinus/AV nodes.

Note: Tiers are illustrative UI indicators, not standardized diagnostic cutoffs.

4. SYSTEM ARCHITECTURE & IMPLEMENTATION

4.1 End-to-End System Topology: CARDIA is implemented as a decoupled two-tier architecture: a Python Flask machine learning inference microservice and a React 18 / TypeScript / WebGL clinical workstation.



4.2 Backend REST API Service: The backend is written in Python using Flask and joblib. Upon initialization, the `ModelRegistry` loads all trained pipelines, feature schema, and background empirical distributions into memory, ensuring sub-50ms inference latency without disk reads. The API exposes five REST endpoints:

- `GET /api/health`: Verifies service liveness, reports all four loaded model pipelines, and returns system timestamp.
- `GET /api/schema`: Serves the complete 54-feature clinical schema, data types, permitted categorical options, and default values.
- `GET /api/metrics`: Returns out-of-fold cross-validation performance metrics, confusion matrices, and selection rationales for all targets.
- `GET /api/sample-patients`: Provides five diverse reference clinical profiles: Low-Risk Asymptomatic, Moderate-Risk Atypical, High-Risk Multi-Vessel Ischemia, Isolated LAD Lesion, and Young Extrapolation Profile (Age 20).
- `POST /api/predict`: Ingests clinical patient JSON; verifies schema conformance; computes multi-target inference probabilities; assigns vessel colormaps; generates local feature explanations; and appends safety disclaimers.

4.3 Strict Input Sanitation & Age Extrapolation Safeguards:

- **Target Quarantine Enforcement:** If an inbound request contains `cath`, `lad`, `lcx`, `rca`, or `cad`, the API immediately halts execution and returns HTTP 400 Bad Request with an explicit anti-leakage diagnostic message.
- **Data Integrity Verification:** Non-numeric characters in numeric fields, categorical values outside permitted sets, `NaN`, and `Infinity` are rejected with HTTP 400.
- **Age Policy & Extrapolation Safeguard:**
  - Ages < 18 or > 100 are rejected with HTTP 400 (`Age must be between 18 and 100`).
  - Ages 18–29 and 87–100 fall outside the observed training cohort (30 ≤ Age ≤ 86 years). The system permits inference for research exploration but returns a prominent clinical extrapolation warning: *"Age is outside the observed training cohort (30–86 years). Prediction reflects statistical extrapolation."*

4.4 Frontend Workstation & 3D WebGL Engine: Built with React 18, TypeScript, Vite, Three.js, and React Three Fiber (`@react-three/fiber`, `@react-three/drei`):

- **BodyParts3D Geometry Integration:** Real anatomical cardiac geometry was sourced from the Database Center for Life Science (DBCLS, Japan) under the CC BY-SA 2.1 Japan license. Anatomical structures—Heart Myocardium (`FMA7088`), LAD (`FMA3862/74912`), LCX (`FMA3895`), and RCA (`FMA50039/3802`)—were centered, oriented to standard Y-up/Z-front coordinates, and exported as native Three.js `BufferGeometry` JSON structures.
- **Interactive Raycasting & Inspection:** Clinicians can independently hover over and select LAD, LCX, and RCA branches. Raycasting identifies the hovered mesh, triggering outline highlights, focal camera centering, and automatic opening of corresponding vessel risk cards.
- **Responsive Clinical Layout:** Styled in pure CSS with a dark clinical aesthetic (near-black background `#050505`, high-contrast typography, charcoal panels). Adapts fluidly across 3-column desktop (≥1200px), 2-column tablet (768–1199px), and single-column mobile (<768px) viewports with zero horizontal scrolling.

5. EVALUATION AND EXPERIMENTAL RESULTS

5.1 Verification Source & Sample Size: All reported evaluation figures derive strictly from out-of-fold cross-validation predictions pooled across all 303 patients (*n* = 303) during Stratified 5-Fold Cross-Validation, serialized in `results/metrics/model_comparison.csv` and `results/metrics/selected_model_metrics.json`.

Table 1: Cross-Validation Model Comparison (*n* = 303, Stratified 5-Fold CV)

TARGET VARIABLE	MODEL ARCHITECTURE	ACCURACY	PRECISION	RECALL	F1-SCORE	ROC-AUC	POS / NEG	DEPLOYMENT STATUS
Cath (CAD) Overall CAD	Logistic Regression	85.15%	92.12%	86.57%	89.26%	92.42%	216 / 87	Evaluated
	Random Forest	**88.12%**	**92.06%**	**91.20%**	**91.63%**	**92.85%**		**Selected**
	Gradient Boosting	83.50%	85.17%	93.06%	88.94%	91.10%		Evaluated
LAD Anterior Descending	Logistic Regression	71.95%	78.05%	72.32%	75.07%	81.54%	177 / 126	Evaluated
	Random Forest	76.57%	79.44%	80.79%	80.11%	84.02%		Evaluated
	Gradient Boosting	**78.55%**	**79.79%**	**84.75%**	**82.19%**	**83.65%**		**Selected**
LCX Circumflex Artery	Logistic Regression	62.38%	51.75%	62.18%	56.49%	66.82%	119 / 184	Evaluated
	Random Forest	**66.34%**	**56.49%**	**62.18%**	**59.20%**	**71.32%**		**Selected**
	Gradient Boosting	66.01%	58.00%	48.74%	52.97%	71.36%		Evaluated
RCA Right Coronary	Logistic Regression	66.34%	54.48%	64.04%	58.87%	70.94%	114 / 189	**Selected**
	Random Forest	66.34%	55.17%	56.14%	55.65%	70.21%		Evaluated
	Gradient Boosting	64.36%	53.41%	41.23%	46.53%	68.84%		Evaluated

Table 2: Deployed Pipeline Performance & Confusion Matrix Breakdown (*n* = 303)

TARGET	SELECTED MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE	ROC-AUC	TN	FP	FN	TP
Cath	Random Forest	88.12%	92.06%	91.20%	91.63%	92.85%	70	17	19	197
LAD	Gradient Boosting	78.55%	79.79%	84.75%	82.19%	83.65%	88	38	27	150
LCX	Random Forest	66.34%	56.49%	62.18%	59.20%	71.32%	127	57	45	74
RCA	Logistic Regression	66.34%	54.48%	64.04%	58.87%	70.94%	128	61	41	73

5.2 Analysis of Target Performance Disparity (LAD vs. LCX/RCA): A critical finding is the pronounced performance gradient across anatomical targets:

- **Strong Discrimination for Cath and LAD:** Overall CAD (ROC-AUC 92.85%, F1 91.63%) and LAD stenosis (ROC-AUC 83.65%, F1 82.19%) demonstrate high discriminative accuracy. The LAD perfuses a large anatomical territory (anterior wall, septum, apex). Severe LAD lesions generate pronounced electrophysiological disruptions visible on standard precordial leads (V1–V4 ST changes, T-wave inversion, Q waves) and resting echocardiography (anterior regional wall motion abnormalities).
- **Attenuated Performance for LCX and RCA:** Prediction for LCX (ROC-AUC 71.32%, F1 59.20%) and RCA (ROC-AUC 70.94%, F1 58.87%) is noticeably weaker. This divergence is grounded in cardiovascular physiology:
  1. *Electrocardiographic Insensitivity:* Standard 12-lead ECG is anatomically insensitive to the lateral and posterior left ventricular myocardium supplied by the LCX. Ischemia in the circumflex distribution is frequently "electrically silent" unless posterior leads (V7–V9) are recorded.
  2. *Coronary Dominance & Collateralization:* In approximately 85% of individuals, the coronary circulation is right-dominant; in ~10% it is left-dominant. A stenosis in a non-dominant LCX or RCA produces minimal resting functional disturbance or biomarker alteration.
  3. *Class Imbalance:* LCX (39.3% positive) and RCA (37.6% positive) are minority classes, providing fewer positive training instances than overall CAD (71.3% positive).

5.3 Methodological Limitations: (1) *Pooled Estimation:* Metrics reflect pooled out-of-fold predictions (*n* = 303); variance across folds is not captured in point estimates. (2) *Selection Optimism:* Selecting the highest-scoring model introduces minor selection optimism. (3) *Calibration Status:* Output probabilities were not formally calibrated with Platt scaling or isotonic regression on a holdout set; they must be interpreted as monotonic risk rankings rather than absolute frequentist probabilities.

6. LIMITATIONS, RESPONSIBLE USE, AND FUTURE WORK

6.1 Explicit System Limitations:

- **Cohort Scale & Diversity (*n* = 303):** Training was conducted on 303 patients from a single tertiary Iranian cardiovascular center. The sample lacks statistical power to represent multi-ethnic genetic backgrounds or rare clinical phenotypes.
- **Tertiary Referral Bias:** All cohort patients presented with clinical indications warranting invasive catheterization (baseline CAD prevalence 71.3%). Deployment in asymptomatic primary care screening populations would increase false positives without model recalibration.
- **Absence of Plaque Localization:** The 3D cardiac model represents idealized reference geometry. CARDIA cannot identify the anatomical segment (proximal, mid, or distal) of stenosis, nor can it characterize plaque morphology, calcification, or ulceration.
- **Associational Nature of Attributions:** Feature importance weights reflect statistical model sensitivities, not causal biological mechanisms. They must not be interpreted as actionable therapeutic targets.
- **Extrapolation Hazards:** Input profiles with age < 30 or > 86 years, or extreme physiological outliers, fall outside the observed training manifold and carry heightened predictive uncertainty.

> **MANDATORY CLINICAL SAFETY NOTICE & RESPONSIBLE USE DECLARATION**
> CARDIA is exclusively an educational research prototype and clinical decision-support proof-of-concept. It is **not** a cleared medical device and must never be utilized as the sole or primary basis for diagnostic, surgical, interventional, or pharmacotherapeutic decisions. All predictions must be evaluated by licensed healthcare professionals in conjunction with standard diagnostic modalities, formal imaging, and patient clinical history.

6.2 Future Work Roadmap:

- **External Multicenter Validation:** Evaluating prospective generalizability across diverse international cohorts (e.g., Cleveland Clinic, Hungarian, and Swiss cohorts) without retraining.
- **Probability Calibration:** Implementing temperature scaling or isotonic regression on holdout validation cohorts to guarantee calibrated predictive probabilities.
- **Native Multimodal Waveform & Video Ingestion:** Transitioning from tabular summaries to native multimodal deep learning architectures capable of directly ingesting raw 12-lead ECG voltage waveforms and echocardiographic video loops.
- **Patient-Specific CCTA Vascular Reconstruction:** Developing deep learning mesh generation pipelines to extract patient-specific 3D coronary geometry directly from DICOM CCTA volumes.

7. REFERENCES AND ATTRIBUTIONS

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